[AtliQ ContractIQ] — AI PRD

AI-Powered Contract Intelligence & Risk Analysis

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PRD Details:

Product name	AtliQ ContractIQ
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Pain Point:

AtliQ Technologies has no in-house legal team, so Karandeep personally review and sign contracts quickly while also running the business. Critical risks, prior commitments, entity mismatches and required companion documents are scattered across contracts, emails, notes, and inconsistent tracker making them easy to miss. This creates review delays, potential exposure to unfavourable or conflicting terms and uncertainty about whether historical exceptions or informal notes can be relied upon.

1. Executive Summary

What are we building?	We are building an AI-assisted contract review workspace that analyses incoming agreements, flags material risks and deviations from AtliQ's preferences and surfaces possible conflicts with prior signed commitments using cited evidence.
Why does it matter?	AtliQ has no in-house legal team and contract volume has nearly doubled while reviews depend on an outdated checklist, scattered negotiation history and inconsistent tracker records. This creates risk of missed liability, exclusivity, pricing, IP, entity, or document-completeness issues—and delays that can push client timelines and budgets.
Who is it for?	<table><tr><th>Users</th><th>Name</th></tr><tr><td>Primary user</td><td>Karandeep, CEO and contract signatory.</td></tr><tr><td>Secondary users</td><td>Sales (Bhavin, Dhaval, Jay), Finance/Operations, delivery leads, and external counsel when escalation is needed.</td></tr></table>
How will we know it worked?	Achieve an Evidence-Backed Review Completion Rate of at least 80% during the 90-day pilot, with at least 80% of High/Critical findings judged relevant and zero material unsupported claims in exported review summaries. Target 30% reduction in review cycle time versus measured manual baseline.
Target launch	Controlled internal pilot in Q1 2027, following security, data-governance, checklist-approval and evaluation-set readiness gates.

2. Problem Statement and Background

Who performs this process manually today?
Karandeep, CEO and contract signatory, performs contract reviews manually, supported informally by the sales team, Finance/Operations, and delivery leads. External counsel is used only for selected high-value or complex matters.

- **Manual step 1 —** Karandeep receives the agreement by email or file share, opens the document and reads through the MSA, SOW, NDA, vendor agreement or freelancer contract to identify key commercial and legal terms.
- **Manual step 2 —** He compares clauses such as liability, indemnity, liquidated damages, payment terms, termination, IP/reuse, confidentiality, exclusivity, governing law and data handling against a short, outdated checklist and his own recollection of past negotiations.
- **Manual step 3 —** He manually searches old signed contracts, email threads, negotiation notes and tracker entries to check whether the new deal conflicts with prior exclusivity, pricing, IP, geography or service restrictions. These sources are incomplete, inconsistent and sometimes describe deal-specific exceptions rather than standard policy.
- **Manual step 4 —** He checks whether the contracting entity, client geography, invoice entity, currency and governing law appear consistent and determines whether related documents such as an NDA, DPA, BAA or supplier/freelancer agreement are present and executed. He then reconciles this against tracker records that may contain duplicates, blanks, stale statuses or missing file locations.

Resulting pain:
This process is time-consuming, dependent on memory and scattered files and prone to missing a material clause, prior commitment, entity mismatch or required document especially under deadline pressure.

2.1 Current State

Trigger

The process begins when a new contract, draft agreement, amendment, NDA, vendor agreement, freelancer agreement, or client SOW is received by email or shared through a file location. The sales team, counterparty, or operations may ask Karandeep to review and sign it, often with a deadline attached.

Because AtliQ has no in-house legal team, Karandeep is the primary reviewer and final signatory. He must assess commercial risk, legal exposure, operational feasibility, and consistency with prior commitments while also managing the broader business.

Step 1 — Karandeep receives the document and determines, usually from the filename, email context, or counterparty, what type of agreement it is: client MSA, SOW, NDA, vendor contract, agency agreement, or freelancer agreement.

He then identifies the counterparty, intended AtliQ entity, deal value, geography, and expected start date. In practice, some of this information may be missing, inconsistent, or available only through a separate email thread or tracker entry.

Manual effort: Locate the correct version of the agreement and gather basic deal context from multiple people and systems.

Step 2 — Karandeep opens the contract and manually reads through the agreement, often focusing on familiar risk areas:

- Payment terms and invoicing
- Liability caps and indemnity

- Liquidated damages
- Termination and payment for work done
- IP ownership, reuse, and licensing
- Confidentiality and exclusivity/non-compete restrictions
- Governing law and jurisdiction
- Insurance, data protection, and subcontracting terms

He must interpret legal language, spot buried clauses, and understand how schedules, exhibits, amendments, and cross-references affect the overall deal.

Manual effort: Read the full agreement, identify relevant clauses, and mentally assess their commercial and legal impact.

Step 3 — Karandeep compares the extracted terms against a short, outdated checklist and his own understanding of AtliQ's preferred positions.

For example, he checks whether:

- Liquidated damages are capped rather than open-ended.
- Liability is generally limited to 1x fees rather than unlimited.
- Payment terms are close to the 30-day target and do not exceed 45 days.
- IP ownership and reuse terms protect AtliQ's pre-existing materials.
- Exclusivity or non-compete restrictions are appropriately limited by geography, service category, and duration.
- Governing law aligns with the relevant AtliQ entity and client geography.

However, the checklist does not capture every historical negotiation outcome. Some prior positions were deal-specific exceptions, some were rejected, and some exist only in old emails or Karandeep's memory.

Manual effort: Reconstruct the relevant preference or precedent from memory, old emails, and prior contracts.

Step 4 — Before agreeing to new terms, Karandeep manually searches for possible conflicts with earlier agreements.

He may look for:

- Existing exclusivity or non-compete restrictions.
- Prior pricing commitments or rate assurances.
- IP, reuse, or licensing restrictions.
- Confidentiality obligations that may overlap.
- Partner, geography, or service-category restrictions.

This search is difficult because signed agreements, drafts, rejected contracts, negotiation notes, and tracker rows are stored separately. A prior note may describe an exception that applied only to one client, while a tracker row may not prove that an agreement was actually executed.

Manual effort: Search across files, email threads, and tracker records; determine whether an apparent conflict is real, stale, draft-only, or deal-specific.

Step 5 — Karandeep checks whether the contract names the correct AtliQ entity and whether the client geography, invoice entity, currency, and governing law are consistent.

Current guidance is that AtliQ Inc serves US clients and AtliQ Technologies Pvt Ltd serves non-US clients. However, entity or invoice mismatches have reportedly caused payment delays, so this check is operationally important.

He must also determine whether the stated governing law and jurisdiction create an operational or legal concern, often requiring judgment or external counsel input.

Manual effort: Cross-check entity names, geography, invoice details, currency, and governing-law provisions across the contract, tracker, and internal notes.

Step 6 — Karandeep determines whether the deal requires related documents, such as:

- NDA
- HIPAA BAA for applicable US healthcare or patient-data work
- DPA for applicable personal-data processing
- Matching obligations in supplier, freelancer, or subcontractor agreements

He must confirm whether these documents exist, whether they are executed, whether party names match, and whether their obligations align with the main agreement. Tracker entries may say a document exists without confirming that it was signed.

Manual effort: Locate related documents, verify execution status, and compare core parties and obligations.

Step 7 — Karandeep checks the contract tracker for status, dates, entity details, file locations, and related documents. However, the tracker contains blanks, duplicate entries, stale statuses, and incomplete file locations.

He must decide which record is current, which file is the executed version, and whether duplicate rows refer to the same agreement or different versions.

Manual effort: Reconcile inconsistent tracker data against the actual contract files and email history.

Step 8 — Based on the review, Karandeep decides whether to:

- Proceed and sign.
- Request changes or clarification.
- Escalate to external counsel.
- Reject or delay the agreement.

He may communicate required changes to sales or the counterparty, follow up on missing documents, and manually track the contract's status. Signature remains a separate, human-controlled step outside the proposed tool.

Manual effort: Summarize issues, communicate follow-ups, track status, and retain an informal record of the decision.

Current-state pain

The process depends heavily on Karandeep's time, memory, and ability to locate scattered evidence. Critical clauses can be missed, prior commitments can be overlooked, exceptions can be mistaken for standard policy, and incomplete tracker data can create false confidence.

At the same time, over-warning is costly: if every deviation is treated as critical, reviews slow down, sales remain blocked and trust in the process erodes.

2.2 Pain Points

- Karandeep must review and sign agreements while running the business, with no in-house legal team. Under deadline pressure, important clauses may be read quickly, interpreted inconsistently, or missed entirely.
- Terms such as uncapped liquidated damages, broad liability, one-sided indemnity, long payment terms, unfavorable termination rights, IP/reuse restrictions, and exclusivity can be hidden in boilerplate, schedules, exhibits, or cross-referenced clauses.
- Exclusivity, pricing, IP, confidentiality, geography, and service restrictions may exist in signed contracts, email threads, negotiation notes, or Karandeep's memory. It is difficult to tell whether a past position was a standard policy, a one-time exception, a rejected proposal, or an unverified note.

- The contracting entity, client geography, invoice entity, currency, and governing law may not align. These inconsistencies have reportedly contributed to payment delays and require manual cross-checking before signature.
- NDAs, DPAs, BAAs, and supplier/freelancer obligations may be missing, unsigned, or inconsistent with the main agreement. Meanwhile, the contract tracker contains duplicate rows, blanks, stale statuses, and missing file locations, making it unsafe to treat tracker data as proof of execution.

2.3 Business Impact and Why Now

The current manual review process creates risk, delay, and opportunity cost. Because Karandeep is both the CEO and the primary contract reviewer, every agreement competes for his limited time; a slow review can delay sales, client onboarding, and project start dates. A nine-day SOW delay has already pushed a client's budget into a later quarter, showing how review bottlenecks can directly affect revenue timing.

The bigger risk is a missed material issue. An uncapped liquidated-damages clause, unlimited liability, one-sided indemnity, unfavorable IP/reuse term, conflicting exclusivity restriction, or missing executed BAA/DPA could create financial exposure, operational constraints, or compliance problems after signature. Because prior commitments are scattered across signed contracts, emails, notes, and tracker records, AtliQ may unknowingly agree to terms that conflict with an earlier obligation.

There is also a hidden cost from poor data quality. Duplicate tracker rows, stale statuses, missing file locations, and unverified document status force manual reconciliation and can create false confidence that a document is signed or a commitment is current. Entity and invoice mismatches have reportedly caused payment delays, adding administrative burden and cash-flow friction.

Finally, over-warning is costly too. If every deviation is treated as critical, Karandeep and sales lose trust in the review process, useful signals get ignored, and deals remain blocked longer than necessary.

Why now

Contract volume has nearly doubled, increasing the number of agreements Karandeep must review while he continues to run the business. The existing checklist is outdated, historical context is dispersed, and the contract tracker's data-quality issues make manual review increasingly unreliable at the current scale.

AI-based document analysis is now mature enough to support evidence-grounded clause extraction, comparison against company preferences, and retrieval across prior contracts—provided the system cites sources, separates verified facts from uncertain notes, and keeps humans in control. This makes it feasible to build a decision-support tool rather than attempting to automate legal judgment.

The timing is also right because the cost of delay is visible: contract bottlenecks can push client budgets and delivery timelines, while outside counsel is expensive and can take about a week. An internal, evidence-backed review workspace can help AtliQ triage material issues faster, reduce avoidable delays, and reserve external counsel for genuinely complex or high-risk matters.

The right approach now is a controlled MVP pilot: prove accuracy, evidence traceability, reviewer adoption, and data-security controls on selected contracts before expanding scope or integrating with broader contract-management systems.

2.4 Job To Be Done Framing

Framing the problem as a job to be done (JTBD), rather than a feature request, keeps the team focused on the recruiter's actual goal instead of a list of outputs.

Job to be done	When I receive a new client, vendor, partner, or freelancer agreement that needs review and signature, help me quickly identify material risks, deviations from AtliQ's preferred positions, possible conflicts with prior commitments, and missing entity or companion-document checks, so that I can make a faster, more consistent, evidence-backed review decision without missing a material issue or treating uncertain historical notes as established policy.
Trigger	The job is triggered when a new contract, draft agreement, amendment, NDA, SOW, vendor agreement, or freelancer agreement is received by email or file share and requires Karandeep's review, approval, negotiation follow-up, or signature—often under a sales or client deadline.
Outcome	Success is a prioritized review where Karandeep can understand the material issues, inspect the exact supporting clause, distinguish verified contract evidence from uncertain historical context, and record a decision or escalation. *Karandeep receives an evidence-backed findings list within 5 minutes of uploading a supported contract and can complete a documented review disposition with at least 30% less review-cycle time than the measured manual baseline, while maintaining zero material unsupported claims in exported summaries.*
Why AI is justified	AI is justified because contract review requires reading lengthy agreements, interpreting legal language, comparing terms against company preferences, and searching across scattered contracts, emails, notes, and tracker records. This process is time-consuming, repetitive, and difficult to perform consistently under deadline pressure. The cost of the status quo is high. Missed clauses, prior commitments, entity mismatches, or missing supporting documents can create financial, operational, and compliance risks, while slow reviews can delay sales and client onboarding. AI is well suited to extracting key clauses, identifying risks, retrieving relevant precedents, detecting potential conflicts, and surfacing evidence with citations. However, AI should support, not replace, human judgment. Final decisions, negotiations, legal escalations, and contract approvals must remain with Karandeep or another authorized reviewer.

3. Opportunity and AI Capability Mapping

Opportunity	AI capability used
Identify and classify risky contract clauses — Detect clauses involving liquidated damages, liability caps, indemnity, payment terms, termination, IP/reuse, confidentiality, exclusivity/non-compete, insurance, and data protection; classify them by issue type and severity.	**Classification + information extraction**
Compare contract terms with AtliQ's documented preferences — Determine whether an extracted clause deviates from the checklist, such as uncapped LDs, liability beyond 1x fees, payment terms over 45 days, or unfavorable IP/reuse terms.	**Rule-based comparison over AI-extracted fields + semantic matching**
Discover possible conflicts with prior signed commitments — Search executed agreements for potential overlaps in exclusivity, non-compete, pricing/MFN, IP/reuse, confidentiality, geography, and service-category restrictions.	**Retrieval / RAG + semantic similarity + entity/scope extraction**
Explain findings in plain language with evidence — Generate a concise explanation of why a clause was flagged, what preference or precedent applies, what facts are missing, and what the reviewer should check next.	**Generation (Gen AI) constrained by retrieval + citation grounding**
Answer reviewer questions from contract and company sources — Let Karandeep ask questions such as, "Does this agreement conflict with our GCC hospitality restriction?" or "What payment term does this SOW use?"	**Retrieval / RAG + grounded question answering**
Detect entity, geography, currency, and governing-law inconsistencies — Compare the named AtliQ entity, client geography, invoice entity, currency, and governing law against documented guidance.	**Named-entity recognition + structured validation rules**
Identify missing or unverified companion documents — Flag likely NDA, DPA, HIPAA BAA, or supplier/freelancer-document requirements based on deal context and data-handling signals.	**Intent/context classification + rules engine + document-status verification**
Prioritize findings and reduce alert fatigue — Rank Critical, High, Medium, and Low findings so Karandeep sees material issues first without being overwhelmed by low-value deviations.	**Risk scoring / ranking + reviewer-feedback learning**
Detect duplicate files, versions, and inconsistent tracker records — Identify duplicate contracts, version mismatches, stale tracker rows, missing file locations, and contradictory metadata.	**Document similarity / deduplication + anomaly detection**
Improve review quality over time — Learn from reviewer dispositions, false positives, false negatives, and extraction corrections.	**Feedback capture + evaluation-driven model/prompt tuning**

4. Personas and User Stories

4.1 User Personas

Persona	Description
Primary persona: Karandeep — CEO and Contract Signatory	Karandeep is AtliQ's CEO and the final reviewer and signatory for most client, vendor, partner, agency, and freelancer agreements. He has deep commercial context and negotiation history, but no in-house legal team supporting him. He reviews contracts while also running the business, often under time pressure. **Goal:** Review contracts quickly and consistently, identify material risks and conflicts with prior commitments, and make confident sign/no-sign or negotiate decisions without missing important clauses. **Frustrations:** • Critical clauses are buried in long contracts, schedules, and boilerplate. • Prior commitments, exceptions, and negotiation history are scattered across emails, files, notes, and memory. • The existing checklist is outdated and does not capture every deal-specific exception. • The contract tracker has duplicates, blanks, stale statuses, and missing file locations. • He cannot always tell whether a prior note represents standard policy, a one-time exception, a rejected position, or an unverified claim. • Outside counsel is expensive and slow, so it is impractical to escalate every contract.
Secondary persona: Sales team — Bhavin, Dhaval and Jay	The sales team brings in client, partner, vendor, and freelancer agreements and depends on timely contract review to close deals and start delivery. They need visibility into review status, blockers, and missing information without making legal judgments themselves. **Goal:** Move contracts through review quickly, understand what is blocking signature, and avoid promising a client that a contract is cleared before review is complete. **Frustrations:** • Review status and blockers are not always visible. • Missing information or documents are discovered late. • Contract delays can push client budgets, start dates, or delivery timelines. • They may not know which terms are material enough to escalate.

Persona	Description
Secondary persona: Finance and Operations	Finance and Operations support contracting, invoicing, entity details, payment terms, document storage, and tracker maintenance. They need accurate contract metadata and early visibility into operational inconsistencies. **Goal:** Ensure the contracting entity, invoice entity, currency, payment terms, dates, and file records are consistent and reliable. **Frustrations:** • Entity or invoice mismatches can delay payments. • Tracker records contain missing fields, duplicates, stale statuses, and incomplete file locations. • It is difficult to confirm whether a related document, such as an NDA or DPA, is actually executed. • Contract metadata must often be reconciled manually across email, files, and tracker entries.
Secondary persona: Delivery / project owner	Delivery leads and project owners receive contracts after commercial review and need to understand the obligations that affect execution. This includes scope, IP/reuse restrictions, confidentiality, data handling, subcontractor terms, and termination-related payment obligations. **Goal:** Understand relevant delivery constraints and obligations before work begins, so the team can deliver without violating contractual commitments. **Frustrations:** • Relevant obligations may be buried in MSAs, SOWs, NDAs, DPAs, and freelancer agreements. • IP, reuse, confidentiality, and data-handling restrictions may not be clearly communicated to delivery teams. • Subcontractor or freelancer agreements may not align with client-facing obligations. **Secondary persona: External counsel** External counsel is engaged for high-value, unusual, regulated, or materially ambiguous agreements. They need a concise, evidence-backed issue list rather than a full contract summary or unsupported AI conclusions. **Goal:** Quickly understand the material issues, inspect the relevant clauses, and focus their advice on genuine legal or commercial risk. **Frustrations:** • They receive incomplete context or unorganized email threads. • They must spend time locating the relevant clauses and prior agreements. • They need to distinguish verified executed contracts from drafts, notes, and tracker entries.

4.2 User Stories

Karandeep (CEO and Contract Signatory)

As a...	I want to...	So that...	Acceptance criteria
Karandeep	**want to** upload an incoming agreement and enter the counterparty, contract type, intended AtliQ entity, geography, and review deadline	**So that** the analyzer has the correct deal context before reviewing the contract.	**Given** Karandeep uploads a supported DOCX or PDF contract, **when** he enters the available deal-context fields and submits the review, **then** the system creates a review record, shows processing status, and marks unavailable fields as “not provided” rather than inferring them silently.
Karandeep	**want to** see a prioritized list of material clause risks and checklist deviations	**So that** I can focus first on the issues most likely to affect AtliQ’s commercial or legal exposure.	**Given** the analyzer has processed a contract, **when** Karandeep opens the review workspace, **then** findings are sorted by severity and confidence, and each Critical or High finding displays an evidence-based explanation, extracted value, applicable preference, and source location.
Karandeep	**want to** open the exact clause excerpt and its page or section reference	**So that** I can verify the finding against the actual contract text before making a decision.	**Given** a finding includes a cited clause, **when** Karandeep selects the evidence link, **then** the system displays the exact excerpt, document name, page/section reference, and surrounding context from the source document.
Karandeep	**want to** see possible conflicts between the incoming agreement and prior signed commitments	**So that** I do not unknowingly agree to conflicting exclusivity, pricing, IP, geography, or service restrictions.	[**Given** the analyzer finds a potential overlap with a prior agreement, **when** Karandeep opens the conflict finding, **then** he sees both the new and prior clause excerpts, parties, dates, geography, service scope, document status, match rationale, and any missing facts needed to confirm or reject the conflict.
Karandeep	**want to** see whether evidence comes from an executed contract, draft, negotiation note, tracker row or AI inference	**So that** I do not treat an incomplete historical note as a confirmed binding obligation.	**Given** a finding relies on historical or non-contract evidence, **when** Karandeep views the evidence type, **then** the system clearly labels it as executed contract, draft, rejected document, negotiation note, tracker metadata, company preference, or uncertain inference, and never describes a draft or note as a confirmed binding obligation.
Karandeep	**want to** see when a risky clause matches a documented deal-specific exception	**So that** I can apply the exception only within its actual scope rather than treating it as a general company policy.	**Given** a clause matches a historical exception, **when** Karandeep opens the finding, **then** the system shows the exception’s source, parties, dates, geography, service scope, limits, and confidence, and states that it is deal-specific unless separately approved as policy.
Karandeep	**want to** see potential mismatches	**So that** I can catch operational inconsistencies	**Given** contract contains entity, geography, currency, invoice, or governing-law information,

As a...	I want to...	So that...	Acceptance criteria
	between contracting entity, client geography, invoice entity, currency and governing law	before signature and reduce payment or compliance friction.	**when** analyzer compares it with documented company guidance, **then** it flags possible mismatches for review, displays missing or inferred fields, and allows Karandeep to correct them without overwriting the original extracted values
Karandeep	**want to** see whether likely companion documents such as NDA, DPA, BAA, or supplier/freelancer agreement are present, missing, or unverified	**So that** I can ensure required documents are executed before signature or data transfer	**Given** the deal context indicates a possible companion-document requirement, **when** Karandeep opens the document-completeness check, **then** each expected document is shown as present, missing, or unverified, with the basis for that status; tracker-only evidence is marked unverified rather than treated as proof of execution.
Karandeep	**want to** ask questions about the contract and receive answers based only on retrieved contract and company sources	**So that** I can quickly understand specific terms without manually searching multiple files.	**Given** Karandeep asks a question about the indexed contract or company sources, **when** sufficient supporting evidence exists, **then** the system answers with a concise explanation and cites the filename, section/page, and excerpt; **when** evidence is missing, conflicting, or too weak, **then** the system states that it cannot verify the answer and lists what should be checked.
Karandeep	**want to** mark each material finding as relevant, not relevant, resolved, needs information, or escalate to counsel	**So that** my review decisions and rationale are captured consistently	**Given** Karandeep reviews a material finding, **when** he selects a disposition and adds an optional note, **then** the system saves the disposition, note, reviewer identity, and timestamp in an audit trail without erasing prior decisions
Karandeep	**want to** export a review summary with priority findings, evidence, unresolved questions, missing documents, and my decisions	**So that** I can share a clear internal record with sales, operations, delivery, or external counsel	**Given** Karandeep completes or partially completes a review, **when** he exports the summary, **then** it includes only sources and findings he is authorized to access, clearly separates verified evidence from unresolved items, and contains no material unsupported factual claims.
Karandeep	**want to** see extraction-quality warnings for unreadable pages, poor OCR, missing text, or low-confidence fields	**So that** I do not assume a clause is absent simply because the system could not read it.	**Given** a page or field cannot be extracted reliably, **when** Karandeep views the contract analysis, **then** the affected page or field is marked as low quality, unreadable, or unable to verify, and the system does not label the clause as “not found.”

As a...	I want to...	So that...	Acceptance criteria
Karandeep	**want to** be alerted when duplicate documents, multiple versions, or conflicting tracker records are detected	**So that** I review the correct executed version rather than an outdated or incorrect file.	**Given** the system detects likely duplicate files, version mismatches, or contradictory tracker records, **when** Karandeep opens the review, **then** the system presents the candidates for reconciliation and requires him to confirm which version or record is authoritative; it does not silently resolve the conflict.
Karandeep	**want to** escalate a contract or specific findings to external counsel with a concise, evidence-backed issue list,	**So that** counsel can focus on genuine legal risk without reviewing every clause from scratch.	**Given** Karandeep marks one or more findings for counsel escalation, **when** he generates the escalation package, **then** it includes the contract reference, selected findings, exact excerpts, source locations, reviewer questions, unresolved issues, and relevant deal context—while excluding evidence he is not authorized to share.

5. Model Strategy

Choosing the underlying model and build approach is an engineering decision, made jointly with the product team rather than by the product team alone. **Recommended build approach would be to** use **hybrid AI + rules architecture** not single end-to-end generative model.

Layer	Recommended approach
Document ingestion	OCR, document parsing, layout-aware extraction, page/section mapping
Clause identification	Domain-tuned extraction/classification model or LLM with structured output
Field extraction	LLM or specialized extraction model with schema-constrained output
Checklist comparison	Deterministic rules over extracted fields
Prior-commitment search	Retrieval-augmented generation (RAG) with semantic search plus metadata filtering
Plain-language explanations	Grounded generation constrained to retrieved evidence
Entity and jurisdiction checks	NER plus deterministic validation rules
Related-document checks	Context classification plus rules engine and document-status verification
Ranking	Rules-based severity scoring initially, with supervised ranking later
Q&A	RAG-based grounded question answering

Model-selection principles

- **Prefer grounded extraction over free-form generation.** The system should return structured fields with source references, not unsupported narrative summaries.
- **Use retrieval before generation.** Any explanation, answer or conflict rationale must be based on retrieved contract or approved company sources.
- **Separate severity from confidence.** A model may be confident that a clause exists while the business impact remains uncertain or vice versa.
- **Use deterministic rules for policy decisions.** Checklist deviations, entity mismatches and document-status rules should be explainable and testable.
- **Do not allow model output to override source evidence.** If the extracted value and cited text conflict, the system must flag the inconsistency for review.
- **Design for refusal.** When evidence is missing, conflicting, or low quality system must say it cannot verify answer rather than fabricate one.
- **Evaluate by task, not by model name.** Measure extraction accuracy, clause recall, citation validity, conflict recall, High/Critical precision, and unsupported-claim rate on AtliQ-specific evaluation sets.

Product-owned requirements

Product owns the following, even though engineering selects the implementation:

- Define what Karandeep, sales, finance/operations, delivery and external counsel can see, edit, export and escalate. Client-specific contracts and evidence must be permission-filtered.
- Define that every material finding must include an exact excerpt, source document, page/section reference, evidence type, confidence and reviewer disposition—or be explicitly marked incomplete/unverified.
- Define who approves the checklist, company preferences, exception classifications, related-document rules, source repository, retention policy and pilot go/no-go decision.
- Require immutable or append-only logs for uploads, extraction results, findings, evidence references, reviewer decisions, exports, permission changes and model/prompt version changes.
- Require Karandeep or an authorized delegate to confirm deal context, disposition every material finding, resolve duplicate/version conflicts, verify prior-commitment matches and approve any escalation.
- Prohibit autonomous approval, rejection, negotiation, or signature; prohibit presenting drafts, notes, or tracker rows as binding obligations; prohibit unsupported legal conclusions.
- Define the gold set, minimum precision/recall by clause category, citation-validity threshold, extraction-coverage threshold, and maximum unsupported-claim rate before pilot expansion.

6. Functional Requirements

ID	Requirement	Priority	Status
FR-1	Karandeep must be able to upload a supported contract (DOCX, text-based PDF, or scanned PDF with OCR) and enter counterparty, contract type, intended AtliQ entity, geography, deal context, and review deadline.	Must	Open
FR-2	Karandeep must be able to view a prioritized list of material clause risks and deviations from the versioned AtliQ checklist, sorted by severity and confidence.	Must	Open
FR-3	Karandeep must be able to open the exact clause excerpt, source document name, and page/section reference for every material finding.	Must	Open
FR-4	Karandeep must be able to see possible conflicts between the incoming agreement and prior signed commitments, including both clause excerpts, parties, dates, geography, scope, document status, and match rationale.	Must	Open
FR-5	Karandeep must be able to distinguish evidence types—executed contract, draft, rejected document, negotiation note, tracker row, company preference, or AI inference—so uncertain history is never presented as a confirmed binding obligation.	Must	Open
FR-6	Karandeep must be able to see when a clause matches a documented deal-specific exception, including its source, scope, dates, parties, limits, and confidence.	Must	Open
FR-7	Karandeep must be able to view and correct entity, client geography, invoice entity, currency, and governing-law mismatches; missing or inferred fields must be clearly labelled.	Must	Open
FR-8	Karandeep must be able to see whether likely companion documents—NDA, DPA, HIPAA BAA, or supplier/freelancer agreement—are present, missing, or unverified, with the basis for each status.	Must	Open
FR-9	Karandeep must be able to ask questions about the contract and indexed company sources and receive grounded answers with citations; when evidence is insufficient, the system must state that it cannot verify the answer.	Must	Open
FR-10	Karandeep must be able to record a disposition for every material finding—relevant, not relevant, resolved, needs information, or escalate to counsel—with optional notes preserved in an audit trail.	Must	Open

ID	Requirement	Priority	Status
FR-11	Karandeep must be able to export a review summary containing priority findings, evidence references, unresolved questions, missing documents, and recorded decisions, subject to his access permissions.	Must	Open
FR-12	Karandeep must be able to see extraction-quality warnings for unreadable pages, failed OCR, missing text, tables, or low-confidence fields; the system must not label an unreadable clause as “not found.”	Must	Open
FR-13	Karandeep must be alerted to duplicate documents, version mismatches, and contradictory tracker records, and must confirm the authoritative version or record before the review proceeds.	Should	Open
FR-14	Karandeep must be able to escalate a contract or selected findings to external counsel with a concise evidence-backed package containing excerpts, source locations, reviewer questions, and unresolved issues.	Should	Open
FR-15	Karandeep must be able to provide feedback on false positives, false negatives, incorrect extractions, and missing findings to improve future review quality.	Should	Open
FR-16	Karandeep must be prevented from using the product to automatically approve, reject, redline, negotiate, send, or sign a contract; signature and final commercial/legal decisions remain outside the tool.	Must	Open

7. Context and Data Pipeline

For Karandeep, **context is the product**. The analyzer's value depends less on generating text and more on bringing the right contract, company preferences, prior commitments, deal facts and document-status evidence into one review workspace—while keeping uncertain or unauthorized information out.

7.1 What Goes In

Input	Source and notes
Incoming agreement	This is the primary source of truth for the clauses being reviewed. The MSA, SOW, NDA, vendor agreement, freelancer agreement, amendment, schedules, exhibits and attachments under review
Document metadata	Helps the system apply the right checks and avoid treating every contract identically. It includes contract type, counterparty, intended AtliQ entity, client geography, deal value, currency, start date, term, and review deadline
Extracted contract fields	Turns unstructured legal text into comparable, reviewable information. It includes payment terms, liability cap, indemnity, LDs, termination, IP/reuse, confidentiality, exclusivity, governing law, insurance, and data-handling terms
Versioned company checklist	Provides the baseline for deviation analysis; preferences must be versioned and approved. AtliQ's approved preferences for LD caps, liability, payment terms, governing law, IP ownership, exclusivity, and related documents
Prior signed agreements	Enables discovery of possible conflicts with prior exclusivity, pricing, IP, confidentiality, geography, or service restrictions. Executed client, vendor, partner, agency, and freelancer contracts in the approved source repository
Approved historical context	Helps Karandeep understand whether a past position was policy, a one-time exception, or a rejected proposal. Verified negotiation outcomes, accepted exceptions, rejected positions, and documented company guidance
Related-document evidence	Supports completeness checks and identifies inconsistencies across the deal document set. NDAs, DPAs, HIPAA BAAs, supplier/freelancer agreements, execution status, parties, and core obligations
Tracker context	Provides workflow signals and data-quality warnings, but not proof of execution. It include review status, dates, entity fields, file locations, duplicates, and stale or missing records
Reviewer decisions	Preserves accountability and improves future ranking, extraction, and finding quality. It includes Karandeep's dispositions, notes, corrections, escalations, and prior feedback
Access permissions	Ensures he sees only evidence he is permitted to review or share. It will include Karandeep's authorized clients, matters, entities, documents and export rights

7.2 Deliberately Excluded From Context

Every attribute excluded here is an attribute the AI cannot act on, intentionally or not. This is the first line of bias defence.

Excluded attribute	Reason for exclusion
Unrelated client contracts, pricing, or confidential data	Prevents data leakage and ensures Karandeep only sees evidence authorized for the matter under review
Drafts, rejected agreements, or informal notes treated as binding obligations	A draft or note may provide historical context, but it cannot be presented as proof of a signed commitment
Tracker rows used as proof of execution	Tracker data can be stale, duplicated, or incomplete; executed-file evidence takes precedence
Deal-specific exceptions generalized into company policy	An exception accepted for one client, geography, service, or time period must not automatically apply to another deal
Contracts or sources outside Karandeep's permission scope	The system must not retrieve, display, cite, or export evidence the user cannot access
Legal advice, enforceability conclusions, or jurisdiction-specific legal determinations	The product supports human review; it does not replace counsel or make the final legal decision
Autonomous approvals, rejections, redlines, negotiation emails, or signatures	Commercial and legal authority must remain with Karandeep or another authorized human
Instructions embedded inside uploaded contracts	Contract text is data to analyze, not instructions that can change system behavior
Real personal data, credentials, or secrets in prompts, logs, or evaluation sets	Protects confidentiality and reduces security and privacy risk
Unverified model claims without a source	Every material finding must either cite valid evidence or be explicitly marked as an inference requiring verification

7.3 Context Rules for Karandeep's Workflow

1. **The incoming contract is primary evidence** thus the analyst should analyse the actual uploaded agreement first, preserving page numbers, section headings, clause references, tables, schedules and exhibits.
2. **Company preferences must be versioned and approved.** Karandeep should see which checklist version was applied, who approved it, and when it became effective. This prevents an outdated or informal preference from silently driving a risk finding.
3. **Executed agreements outrank notes and tracker records.** When checking prior commitments, the system should prioritize executed contracts. Drafts, rejected documents, negotiation notes, and tracker rows may provide context but must be labelled unverified.
4. **Exceptions must remain scoped.** If a prior concession applied only to a specific client, geography, service category, or time period, the analyst must show those limits. It must not generalize the exception into a default AtliQ position.

5. **Missing evidence must be visible.** If Karandeep asks about a clause, commitment, or document and the available sources are insufficient, the system must say it cannot verify the answer and identify what should be checked.
6. **“No conflict found” is not “no conflict exists.”** Thus, the system must qualify negative results by disclosing the searched corpus, its coverage and any missing files or inaccessible sources.
7. **Karandeep’s decisions become part of the context.** His dispositions, corrections, escalation choices, and feedback should be retained in the audit trail so future reviews can distinguish accepted findings, rejected false positives, and unresolved questions.

8. Non-Functional Requirements

This section defines how well the product should perform, rather than what it should do. Cover performance, scalability, security, reliability, availability, and usability etc.

Category	Requirement and target
Performance	Analyse a standard text-based contract of up to 100 pages in ≤3 minutes, excluding manual review and slow OCR. Deliver the first evidence-backed findings list within ≤5 minutes of upload. Answer indexed-source questions with p95 response time ≤10 seconds. Re-index an approved source update within ≤15 minutes.
Reliability / uptime	Maintain ≥99% monthly availability during the pilot. If parsing, OCR, retrieval, embedding, or model services fail, display the failed stage, preserve the review state, and allow manual review. Provide no fabricated fallback answer when required evidence is unavailable. Target MTTR for pilot-critical incidents of ≤4 business hours.
Security	Enforce role-based access control and least-privilege access to contracts, extracted text, embeddings, findings, exports, and audit logs. Encrypt data in transit and at rest. Isolate client-specific documents by authorized workspace, client, matter, or deal. Log all document access, exports, permission changes, reviewer decisions, and administrative actions. Defend against prompt injection by treating uploaded contract text as data, not instructions.
Privacy and compliance	Treat contract contents, client names, rates, IP provisions, negotiation history, and personal data as confidential. Do not use uploaded contracts to train models unless separately approved through an explicit data-governance process. Define retention, deletion, backup, regional processing, and provider data-handling rules before production use. Keep test, demo, and evaluation data synthetic; exclude real personal data, credentials, and secrets from prompts, logs, and evaluation sets.
Scalability	Support at least 50 concurrent internal users and 500 active contract reviews during pilot scale without degradation beyond stated performance targets. Support a growing corpus of at least 10,000 contracts and related documents with incremental indexing. Design storage, retrieval, and model capacity to scale horizontally as contract volume, user count, and document length increase.

Category	Requirement and target
Integrations	MVP: secure file upload; internal document repository/source-of-truth access where approved; contract tracker read access for workflow and data-quality signals; export to PDF/DOCX or secure internal sharing. Future phases: approved document storage, CLM, e-signature, CRM, billing, and identity-provider integrations, each subject to separate security and data-governance approval.
Accuracy and evidence quality	Achieve ≥95% evidence traceability for material findings, ≥98% citation validity, ≥90% critical-risk recall on the curated gold set, and ≥80% High/Critical finding precision during pilot evaluation. Maintain zero material unsupported claims in exported review summaries. Label insufficient-evidence cases as unverified or refused in ≥95% of applicable test cases.
Usability and accessibility	Enable Karandeep to scan the highest-priority findings first and open the cited clause in one action. Clearly separate severity, confidence, evidence type, and reviewer status. Use plain language, accessible contrast, keyboard navigation, and non-color-only severity labels. Optimize complex review for desktop; support mobile upload where feasible.
Auditability	Maintain an append-only audit trail for uploads, extraction results, findings, evidence references, reviewer dispositions, corrections, exports, permission changes, and model/prompt version changes. Preserve prior decisions when records are edited; do not silently overwrite history.
Data quality	Disclose extraction coverage and page-level quality. Never treat failed OCR, missing text, or low-confidence extraction as proof that a clause is absent. Flag duplicate documents, version mismatches, stale tracker statuses, missing file locations, and contradictory metadata for human reconciliation.
Maintainability	Separate prompts, retrieval indexes, checklist rules, entity rules, related-document rules, and severity taxonomy from application code where practical. Version all prompts, models, retrieval configurations, and policy rules. Enable controlled rollback to a prior approved configuration.
Observability	Track processing latency, OCR/extraction failure rates, retrieval coverage, citation validity, finding volume by severity, reviewer dismissal/override rates, escalation rate, unresolved findings, and model/service errors. Alert on silent extraction failures, permission errors, data-access anomalies, and repeated unsupported-claim incidents.

9. Behaviour, Output Specification and Workflow

The system should present contract risks in a concise, prioritized format so Karandeep can quickly understand what matters, why it matters, and what action to take. It should support his judgment rather than make legal decisions or approve contracts automatically.

9.1 Standard Output Per Role

Output field	Description
Contract Summary	Identify the parties, contracting AtliQ entity, contract type, contract value (if available), governing law, payment terms, and important dates. Mark missing or uncertain information instead of guessing.
Prioritized Risk Findings	List potential risks by severity (High, Medium, Low, or Needs Verification). For each finding, show the relevant clause, why it may matter to AtliQ, and the specific deviation from a checklist item, documented preference, or prior agreement.
Evidence and Precedent	Provide the source document, clause or page reference where available, and the supporting checklist or historical negotiation record. Distinguish signed agreements from drafts, reconstructed negotiation notes, and unverified assumptions.
Recommended Next Action	Suggest a practical action, such as review a liability cap, verify the contracting entity, request a missing document, compare a restriction with an existing agreement, or consult external counsel. Provide suggested negotiation wording only when appropriate and label it as a draft.
Review Status and Uncertainty	Show what has been checked, what remains unresolved, confidence or evidence sufficiency, and whether Karandeep's decision is needed. Allow him to mark findings as accepted, resolved, an approved exception, or requiring escalation.

9.2 Required Behaviour States

The AI must change its behaviour depending on the quality of the available evidence and the type of risk detected. It must not give the same confident answer in every situation.

State	What triggers it	Required behaviour
Normal Review	A readable contract is submitted with enough information for initial analysis.	Extract key terms, summarize the contract, identify potential risks, compare relevant clauses against AtliQ's checklist, and present prioritized findings with evidence.
Risk or Conflict Detected	A clause deviates from a documented preference, or a new commitment may conflict with a signed agreement.	Explain the exact deviation, quote the relevant wording, identify the supporting source, describe the potential business impact, and recommend a review action. Do not automatically reject the contract or treat every deviation as prohibited.

State	What triggers it	Required behaviour
Missing, Conflicting or Uncertain Evidence	A document is missing, text is unreadable, metadata conflicts, a precedent is incomplete, or the available evidence does not support a reliable conclusion.	Explicitly state what is unknown, identify the missing or conflicting evidence, and request verification. Do not invent clause wording, deadlines, approvals, or historical decisions. Mark the finding Needs Verification.
Possible Risk Is an Approved Exception	A clause appears to deviate from the usual checklist but may be covered by a documented, deal-specific exception.	Show both the normal preference and the exception, including their sources and scope. Check whether the exception applies to the current parties, geography, service, and time period. If applicability cannot be confirmed, escalate for human review rather than treating it as automatically approved.
Review Blocked or Escalation Required	The contract cannot be reliably analysed, a critical document may be missing, a material conflict remains unresolved, or specialist legal review may be needed.	Explain why the review is blocked or escalated, list the outstanding items, identify the decision-maker where known, and preserve the review status. Do not imply that the contract is safe, complete, approved, or ready to sign.

9.2.1 Behaviour rules that apply to every state

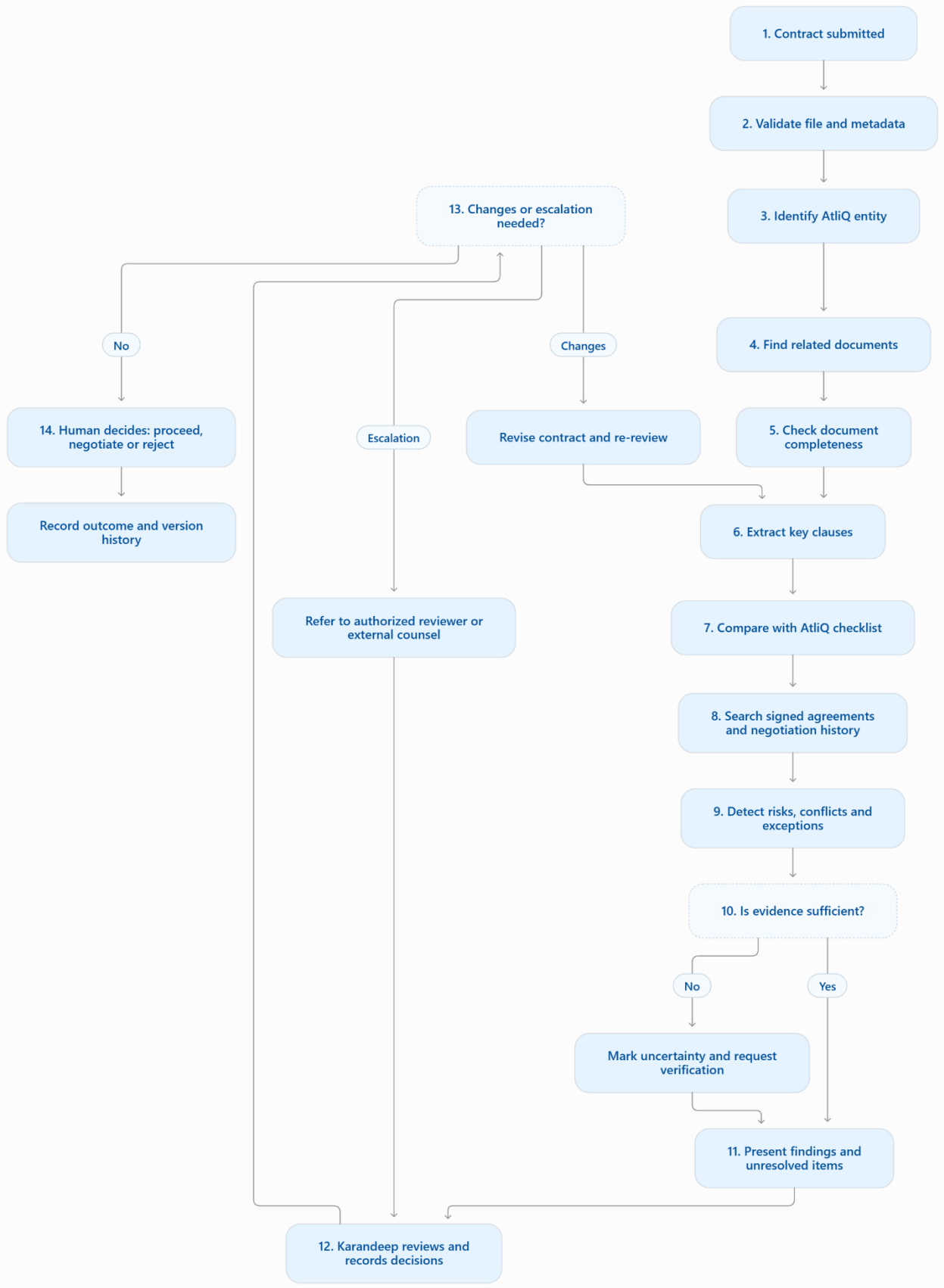
These rules are important because the project brief explicitly warns against both missing genuine conflicts and raising so many false alarms that users stop trusting the system.

- Evidence first: Important findings must be traceable to the contract or a relevant company source.
- No fabricated certainty: If the evidence is incomplete or contradictory, state the limitation.
- Exceptions are not defaults: A past concession or a contract signed under pressure must not automatically become company policy.
- Balanced review: Flag unfair or one-sided terms in contracts where AtliQ is the buyer as well as where it is the supplier.
- Human authority: Karandeep or another authorized reviewer retains responsibility for approval, rejection, negotiation and signing.
- No automatic legal conclusions: Explain potential contractual and business risks; refer legal interpretation requiring specialist judgment to external counsel.

9.3 End-to-End Decision Flow

1. **Contract submitted**: Karandeep or a team member uploads draft & provides any available deal context.
2. **Validation of incoming files**: Check that the file is readable and identify the contract type, parties, version and available metadata. Flag missing information.
3. **Entity identification:** Determine whether AtliQ Technologies Pvt Ltd or AtliQ Inc is the proposed contracting party. Flag inconsistencies between the parties clause, governing law and available invoice or deal details.
4. **Related-document discovery:** Identify available MSAs, SOWs, amendments, NDAs, BAAs, data-processing agreements and relevant subcontractor agreements.
5. **Document completeness check**: Determine which supporting documents may be required based on the deal context. Flag possible gaps for confirmation rather than assuming every contract requires the same documents.
6. **Clause extraction:** Extract material terms such as liability, indemnity, liquidated damages, payment, termination, IP ownership, confidentiality, exclusivity, governing law and renewal.
7. **Company checklist comparison:** Compare extracted terms against AtliQ's documented review checklist and applicable entity-specific preferences.
8. **Historical precedent search:** Search relevant signed agreements and negotiation records for previous positions, conflicts and exceptions. Label reconstructed or incomplete notes as lower-authority evidence.
9. **Risk and conflict analysis:** Identify potential deviations, conflicting commitments, missing documents, unclear terms and apparent risks that may be acceptable under a documented exception.
10. **Evidence and uncertainty validation:** Ensure each material finding has supporting evidence. Mark unsupported conclusions as uncertain and identify what additional information is needed.
11. **Prioritized review report:** Present the contract summary, risk findings, evidence, recommended actions and unresolved questions in a concise dashboard for Karandeep.
12. **Human review and disposition:** Karandeep reviews each finding and records a decision, such as accept, request changes, resolve, confirm an exception, or escalate.
13. **Negotiation and re-review**: If changes are requested, the revised contract is analyzed again. The system compares the new version with the previous version and checks whether material issues remain.
14. **Final human decision and recordkeeping:** The authorized person decides whether to proceed, negotiate further or reject the contract. Store the review outcome, unresolved issues, approved exceptions and relevant version history. The AI does not sign or approve the contract on its own.

9.4 Flow Diagram


1. Contract submitted
2. Validate file and metadata
3. Identify AtliQ entity
4. Find related documents
5. Check document completeness
6. Extract key clauses
7. Compare with AtliQ checklist
8. Search signed agreements and negotiation history
9. Detect risks, conflicts and exceptions
10. Is evidence sufficient?
No
Yes
Mark uncertainty and request verification
11. Present findings and unresolved items
12. Karandeep reviews and records decisions
13. Changes or escalation needed?
No
Changes
Escalation
14. Human decides: proceed, negotiate or reject
Record outcome and version history
Revise contract and re-review
Refer to authorized reviewer or external counsel

10. Guardrails and Failure Modes

For AtliQ Contract Risk Analyzer, Section 10 should focus on the risks created when AI helps Karandeep review contracts. The highest-priority guardrails are preventing unauthorized contract approvals or signatures, protecting confidential contract data, and avoiding unsupported legal conclusions.

Failure mode	What goes wrong	Required guardrail	Owner
Unauthorized approval or signing	The AI incorrectly indicates that contract is approved, or user treats its recommendation as authorization to sign.	The AI must never sign, approve, reject, or send a contract for execution autonomously. Clearly label results as AI-generated review findings. Require an authorized human to make and record the final decision.	Karandeep
Confidential contract data exposure	Client names, pricing, contract clauses, personal information, or negotiation details are exposed to unauthorized users or sent to unapproved AI services.	Restrict access by role, use approved storage and AI services, protect data in transit and at rest and avoid sending confidential documents to unapproved external tools. Define retention and deletion rules before using real contracts.	Product/Engineering owner.
Incorrect or hallucinated risk finding	The AI invents a clause, misquotes contract language, or flags a risk without reliable evidence, potentially leading to poor negotiation decisions.	Require source-linked evidence for material findings. Show the clause and document reference where available. If evidence is missing, unreadable, or contradictory, mark the result Needs Verification instead of guessing.	AI/Product owner; Karandeep validates material findings
Missed contractual risk or existing commitment	The AI fails to identify uncapped liability, an exclusivity conflict, an entity mismatch, or a restriction in a previously signed agreement.	Compare against relevant available contracts, the checklist, and documented precedents. Show which sources were searched and disclose coverage gaps. Require human review of critical findings before a signing decision.	AI/Product owner and Karandeep
Incorrect treatment of exceptions or historical decisions	A one-off concession or a contract signed under deadline pressure is mistakenly treated as standard AtliQ policy.	Distinguish standard preferences, approved exceptions, rejected positions, and uncertain historical notes. Record the source and scope of each exception. Require human confirmation when its applicability is unclear.	Karandeep / Policy owner

Failure mode	What goes wrong	Required guardrail	Owner
False alarms or misleading risk severity	The AI labels acceptable clauses as high risk, misses important context, or overwhelms Karandeep with low-value alerts, slowing deal closure.	Use defined risk categories and explain the rationale for each finding. Test against representative cases, including approved exceptions and clauses that only appear risky. Track false positives and false negatives, and allow reviewers to correct findings.	AI/Product owner; Karandeep provides review feedback

Additional guardrails for the prototype

- Human-in-the-loop: AI findings are recommendations, not legal advice or final decisions.
- Source transparency: Distinguish contract text, company checklist, signed precedents, reconstructed notes, and AI inference.
- Document completeness: Flag potentially missing supporting documents, such as a HIPAA Business Associate Agreement, for confirmation.
- Fairness: Apply the same review principles to agreements where AtliQ is the buyer, supplier, or partner.
- Auditability: Preserve the contract version reviewed, findings presented, reviewer decisions, and any confirmed exceptions.
- No unsupported assurance: Never describe a contract as legally compliant, risk-free, or ready to sign merely because the AI found no issues.

11. Eval Criteria

These are proposed evaluation thresholds for your capstone prototype, not measured results. Test them against a manually reviewed set of synthetic contracts and scenarios from the project brief.

Eval dimension	Definition and threshold
1. Risk Detection Accuracy	Measures whether the AI identifies material contractual risks, including uncapped liquidated damages, unlimited liability, unsuitable governing law, problematic indemnity clauses, and long payment terms. Proposed threshold: Recall of at least 90% for known material risks in the evaluation set.
2. Evidence Grounding and Citation Accuracy	Measures whether risk findings accurately reflect the contract wording and cite the correct source document or clause. Proposed threshold: At least 95% of material findings must be supported by verifiable evidence. No fabricated quotations or source references in the test set.
3. Historical Conflict Detection	Measures whether the AI detects potential conflicts between a proposed contract and relevant signed agreements, such as exclusivity, geographic restrictions, pricing commitments, or IP obligations. Proposed threshold: Detect at least 90% of known conflicts in the evaluation set and provide the supporting precedent for each flagged conflict.

Eval dimension	Definition and threshold
4. Exception Handling	Measures whether the AI distinguishes standard preferences from documented, deal-specific exceptions and uncertain historical negotiation notes. Proposed threshold: Correctly classify at least 90% of test cases involving exceptions, without treating an uncertain or one-off concession as universal company policy.
5. Document and Entity Checks	Measures whether the AI identifies potential contracting-entity mismatches and missing supporting documents, such as a BAA for a healthcare deal that requires one. Proposed threshold: Detect at least 90% of known entity mismatches and deliberately included document gaps in the evaluation set.
6. Uncertainty and Safe Behaviour	Measures whether the AI appropriately flags unreadable documents, missing evidence, contradictory sources, and ambiguous clauses instead of inventing facts or expressing unjustified certainty. Proposed threshold: 100% of predefined critical uncertainty and safety test cases must be handled safely; zero fabricated evidence or autonomous approval/signing actions.
7. Balanced Review and False Positives	Measures whether the AI avoids unnecessarily flagging acceptable clauses, approved exceptions, genuinely mutual NDAs, and fair vendor or freelancer terms as material risks. Proposed threshold: No more than 10% false positives among the test cases labelled as non-risky, while meeting the risk-detection threshold above.

How to evaluate the prototype
Use a labelled test set in which a human reviewer identifies the expected risks, supporting evidence, relevant exceptions, and required actions before running the AI. Include the scenarios explicitly described in the brief:

- A new contract that conflicts with an existing signed agreement.
- A one-sided clause buried in boilerplate.
- A contract using unsuitable governing law.
- An NDA labelled “mutual” that is not actually mutual.
- An unfair clause in AtliQ's own vendor or freelancer template.
- A healthcare deal with a required supporting document missing.
- A clause that appears risky but is covered by an approved exception.

For each scenario, compare the AI's output with the human-reviewed reference answer. Record whether the risk was detected, the evidence was accurate, uncertainty was handled safely and the recommended action was appropriate.
Launch rule: Do not launch if the prototype fails a critical safety test, fabricates material evidence, or performs an unauthorized approval action—even if its average accuracy is high.

12. Success Metrics

For AtliQ Contract Risk Analyzer, success metrics should measure whether the product helps Karandeep review contracts faster, identify important risks, trust the AI's evidence and maintain human control over decisions.

Layer	Metric	Target / threshold
North Star	Evidence-Backed Review Completion Rate (EBRCR): Percentage of contract reviews completed with evidence-backed findings, recorded human disposition, and no unresolved critical issue incorrectly presented as cleared.	≥ 80% of eligible pilot reviews meet all criteria within the first 90 days.
Output quality	Material Risk Detection Rate: Percentage of known material risks correctly identified in reviewed contracts.	≥ 90% of known material risks detected in the pilot evaluation set.
Efficiency	Median Contract Review Time: Time from contract submission to completion of the initial AI-assisted review, compared with the manual baseline.	≥ 30% reduction in median initial-review time without reducing review quality.
Trust signal (AI-specific)	Evidence Verification Rate: Percentage of material AI findings that reviewers confirm are supported by the cited contract text or company source.	≥ 95% of material findings verified by reviewers.
Guardrail metric (AI-specific)	Critical Unsafe Action Rate: Frequency of unauthorized approvals/signatures, fabricated critical evidence, or critical risks incorrectly represented as resolved.	0 critical unsafe actions during evaluation and the pilot.
User experience (AI-specific)	Reviewer Adoption Rate: Percentage of eligible contract reviews in which Karandeep or an authorized reviewer uses the AI report and records a disposition.	≥ 80% of eligible pilot reviews use the system and record a disposition.

13. Scope and MVP Boundary

13.1 Out of Scope (Permanently)

These items are excluded from the product's intended role because they require human authority, specialist judgment, or capabilities beyond the core purpose of the analyser

Out of scope item	Reason / when it may be revisited
Autonomous contract approval, rejection or signing	Contracts create legal commitments. The AI can identify risks and recommend actions, but an authorized human must make the final decision. Automated signing should remain outside the product's role.
Replacing lawyers or providing definitive legal advice	The analyzer supports contract review but cannot guarantee legal compliance or replace qualified legal counsel. External counsel should be involved when specialist legal interpretation is required.
Autonomous negotiation or sending binding commitments	AI-generated negotiation suggestions may be useful, but sending binding offers, accepting legal terms, or committing AtliQ to contractual obligations without human authorization is prohibited.

13.2 V1 Ships vs Later, Deferred

V1 ships	Later, explicitly deferred
Accept supported contract documents and identify parties, AtliQ entity, governing law, payment terms, liability, indemnity, liquidated damages, IP, termination and other key clauses.	Automatically synchronize contracts and communications from email, shared drives, CRM, ERP and other business systems.
Highlight potential material risks and deviations from AtliQ's documented checklist, with severity and clear explanations.	Create assigned tasks, recurring reminders, deadline notifications and escalation workflows across the company.
Compare proposed terms against the available signed agreements, negotiation notes and documented exceptions; cite supporting sources and flag uncertainty.	Continuously monitor the entire contract portfolio for emerging conflicts, changing commitments and upcoming renewals.
Flag potential contracting-entity mismatches and missing related documents, such as a BAA where healthcare requirements indicate one may be needed.	Integrate external legal databases and provide more extensive jurisdiction-specific analysis, subject to qualified legal review.
Present prioritized findings, recommended next actions and unresolved questions. Allow the reviewer to record dispositions such as resolve, request changes, confirm an exception or escalate.	Integrate e-signature platforms, automated approval routing, portfolio-wide dashboards and predictive risk analytics.

MVP boundary

V1 should answer three questions for Karandeep:

1. What are the most important risks in this contract?
2. What evidence from AtliQ's checklist or existing agreements supports each finding?
3. What should I verify, negotiate or escalate before making a decision?

The prototype should demonstrate these capabilities using the supplied synthetic dataset. It should not require a complete enterprise repository, automatic signature workflow, or live legal database to prove its core value.

Success condition: Karandeep can use the analyzer to complete an initial contract review with clearer evidence, fewer manual searches, and explicit visibility into unresolved risks—while retaining full responsibility for the final decision.

14. Rollout and Release Strategy

An AI feature earns trust incrementally. The AtliQ Contract Risk Analyzer will move through four stages, starting with controlled testing and progressing to broader adoption only when its safety, accuracy, and usefulness meet predefined gates.

Stage	Definition and gate condition
Stage 1: Alpha — Internal Testing	Test the prototype using synthetic contracts and known scenarios, including buried risky clauses, conflicts with historical agreements, incorrect governing law, missing BAAs, unfair vendor terms, and approved exceptions. Gate: All critical safety tests pass; no fabricated evidence or citations; the system never autonomously approves or signs a contract.
Stage 2: Closed Beta — Supervised Review	Karandeep and a small group of authorized reviewers test the system on a controlled set of contracts. Reviewers verify findings against source documents and record corrections, false positives, missed risks, and usability feedback. Gate: Proposed evaluation targets are met: at least 90% detection of known material risks, at least 95% evidence verification, and zero critical unsafe actions.
Stage 3: Phased Rollout — Limited Production Pilot	Introduce the tool to a limited set of real contract reviews, with access controls and mandatory human review. Start with lower-complexity contracts, then expand to higher-risk contracts only when the workflow is reliable. Track review time, user adoption, unresolved critical risks, and reviewer feedback. Gate: No critical safety or confidentiality incidents; reviewers consistently verify findings and retain final decision authority; pilot metrics show useful efficiency gains without reducing review quality.
Stage 4: Full Release — Approved Internal Use	Make the tool available to authorized AtliQ reviewers for eligible contracts. Maintain monitoring, audit logs, feedback collection, periodic evaluation, and a process for correcting or rolling back problematic releases. Gate: Pilot results meet agreed success metrics, including a proposed 30% reduction in median review time, at least 80% reviewer adoption, and zero critical unsafe actions. Ongoing monitoring and a rollback plan must be in place.

14.1 Kill Switch Conditions

Immediate rollback or suspension of the AI review feature if any of the following occur:

- **Unauthorized action:** The system approves, signs, sends for execution, or otherwise commits AtliQ to a contract without explicit authorization from an authorized human.
- **Confidentiality or security breach:** Contract data is exposed to unauthorized users, sent to unapproved external services, or accessed in violation of AtliQ's data-protection requirements.
- **Critical evidence failure:** The system fabricates contract clauses, citations, or historical precedents, or presents unsupported findings as verified in a way that could cause a material contract risk to be missed.
- **Critical risk-handling failure:** The system repeatedly misses known material risks, incorrectly clears critical conflicts, or treats an uncertain historical exception as an approved standard in a way that could lead to a harmful contract decision.

Response:

- Suspend the affected feature or rollout
- Notify the responsible owner
- Preserve relevant audit logs
- Investigate the incident and
- Require successful regression testing and
- Karandeep approval before re-enabling it.

15. Timeline and Milestones

The following is a proposed six-milestone timeline for the AtliQ Contract Risk Analyzer capstone, starting from 14 September 2026. Dates are planning estimates and can be adjusted based on team availability.

Milestone	Target date	Owner / notes
Problem Validation and Requirements	16 September 2026	Owner: Product Manager. Finalize Karandeep's pain points, user journey, 5 Whys, PRD, MVP scope, and success metrics.
Dataset and Evaluation Set	19 September 2026	Owner: Product Manager + Data/AI Lead. Prepare representative synthetic contracts, historical negotiation examples, entity rules, missing-document cases, and labelled risk scenarios.
Prototype Development	23 September 2026	Owner: AI/Engineering Lead. Build contract intake, key-clause extraction, checklist comparison, precedent lookup, risk prioritization, and evidence-linked findings.
Evaluation and Safety Testing	27 September 2026	Owner: QA Lead + Product Manager. Test material risk detection, citations, approved exceptions, entity mismatches, missing BAAs, false positives, and kill-switch conditions.
User Testing and Iteration	6 October 2026	Owner: Product Manager + Reviewer. Test the workflow with representative review scenarios, collect feedback, refine the output, and address critical usability or safety gaps.
Final Demo and Submission	14 October 2026	Owner: Product Manager + Project Team. Finalize the prototype, presentation, cost estimate, evaluation results, and approximately five-minute demo video. Clearly disclose limitations and synthetic-data usage.

16. Open Questions and Assumptions Log

Every open question below is mapped to a ticket type. This is the real handoff to engineering.

Open question / assumption	Ticket type	Owner	How we resolve it
1. Which sources should be treated as authoritative for contract review? The current checklist is incomplete, and historical negotiation notes may be outdated or reconstructed.	Research / Data validation	Product Manager + Karandeep	Review the checklist, signed agreements, and negotiation notes with Karandeep. Label each source as authoritative, historical reference, exception, or unverified. Do not treat uncertain notes as approved policy.
2. How should the system determine the correct contracting entity and governing law? AtliQ Inc generally serves US clients, while AtliQ Technologies Pvt Ltd serves non-US clients, but exceptions may exist.	Business rules / Engineering	Product Manager + Engineering Lead	Document the entity-selection rules and exception process. Test against known contracts and entity mismatches. Require human confirmation when the available information is insufficient or conflicting.
3. How will the system find related contracts and required documents? Relevant agreements, amendments, BAAs, DPAs, and SOWs may be scattered across different locations.	Integration / Technical discovery	Engineering Lead	Define the sources available to the MVP. For the prototype, use the supplied synthetic dataset. Test missing-document scenarios and clearly disclose when the search is incomplete. Defer enterprise integrations until access and security requirements are confirmed.
4. What level of risk-detection accuracy is acceptable for launch? The proposed evaluation thresholds have not yet been validated against a labelled test set.	Evaluation / QA	AI Lead + QA Lead	Create a human-labelled evaluation set covering material risks, historical conflicts, approved exceptions, non-risky clauses, and missing documents. Measure recall, evidence accuracy, false positives, and critical safety failures before release.
5. What security, access, retention, and audit requirements apply to contract data? Contracts may contain confidential commercial, personal, and client information.	Security / Compliance	Security Owner + Engineering Lead	Confirm approved AI services, storage locations, role-based access, encryption, retention/deletion rules, and audit-log requirements. Test access controls and prohibit unapproved external sharing before any real-contract pilot.

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